

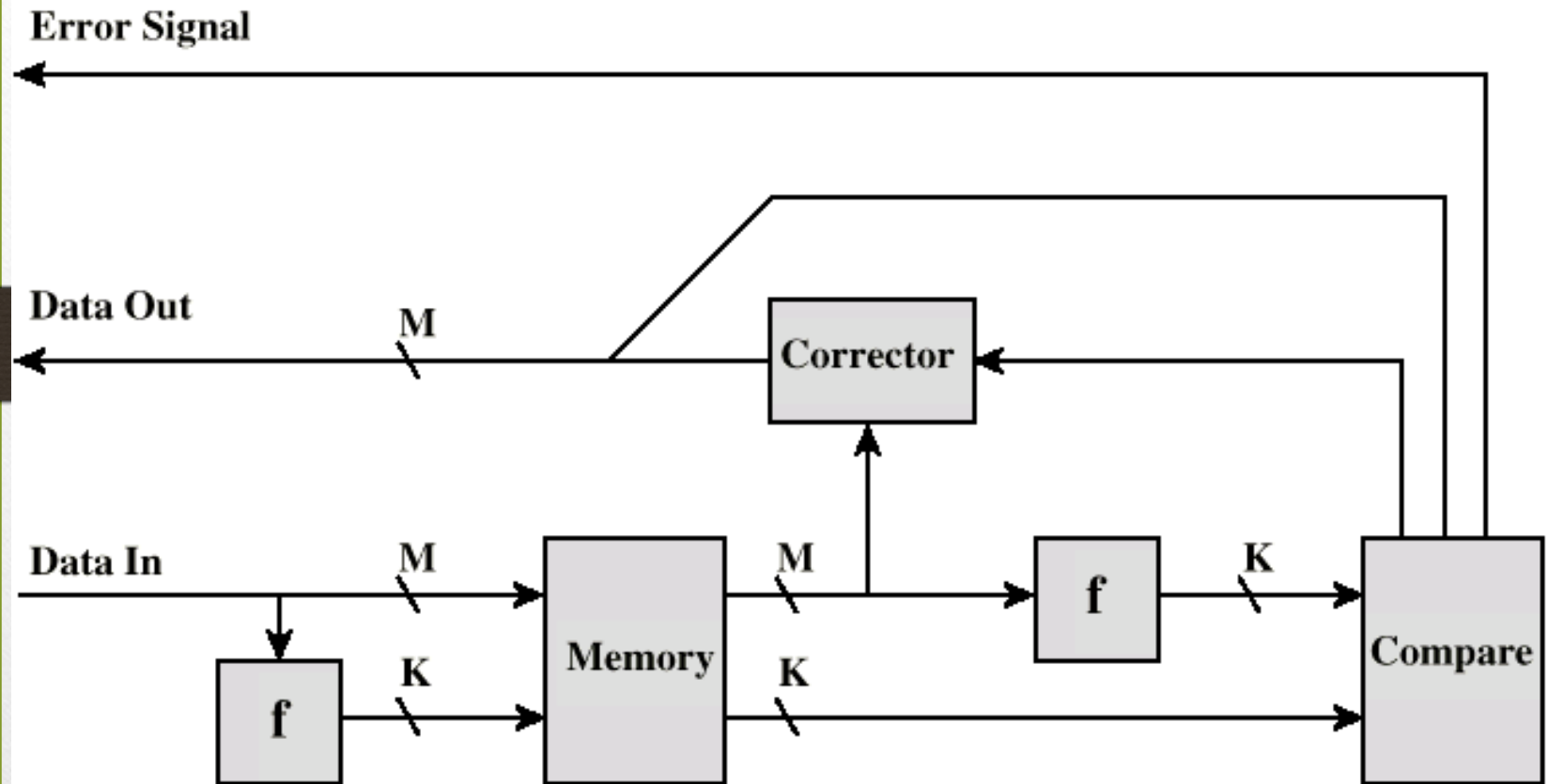
Error correction and detection

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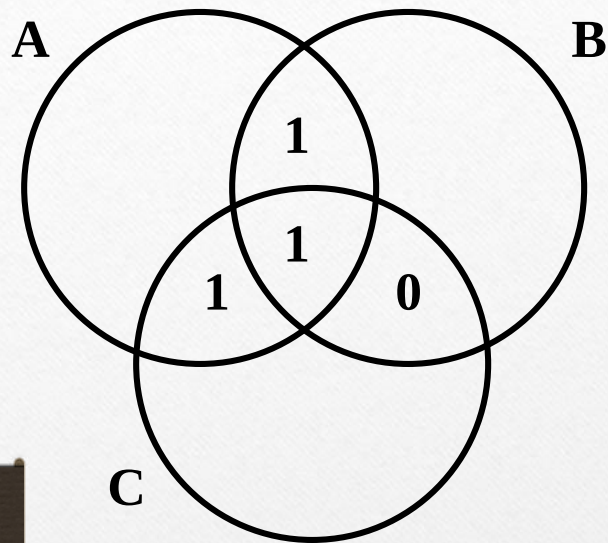
Error Correction

- A semiconductor memory system is subject to errors.
- Hard failures – permanent physical defects
Environmental abuse, manufacturing defects, etc.
- Soft error
Power supply problems etc.
- Need logic for detecting and correcting errors.
- Basic technique
 - Prior to storing data a code is generated from the bits in the word.
 - Code is stored along with the word in memory.
 - Code used to identify and correct errors.
 - When the word is fetched a new code is generated and compared to the stored code.
 - No error (normal case)
 - Correctable error is detected and corrected.
 - Non-fixable error is detected and reported.

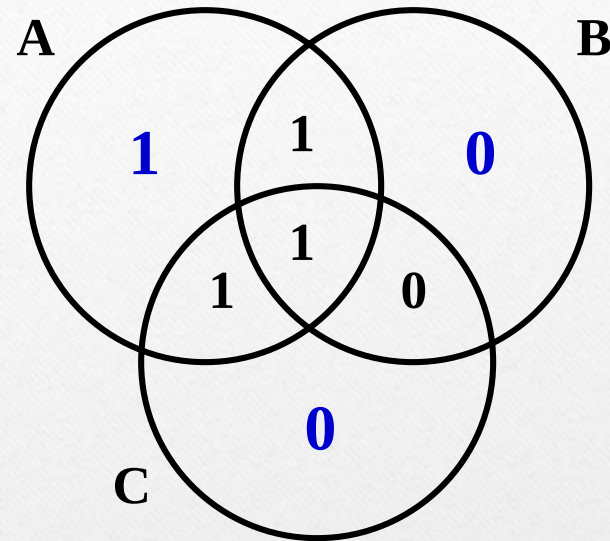
Error Correcting Code Function



Hamming Code



Assign data bits to the inner compartments.

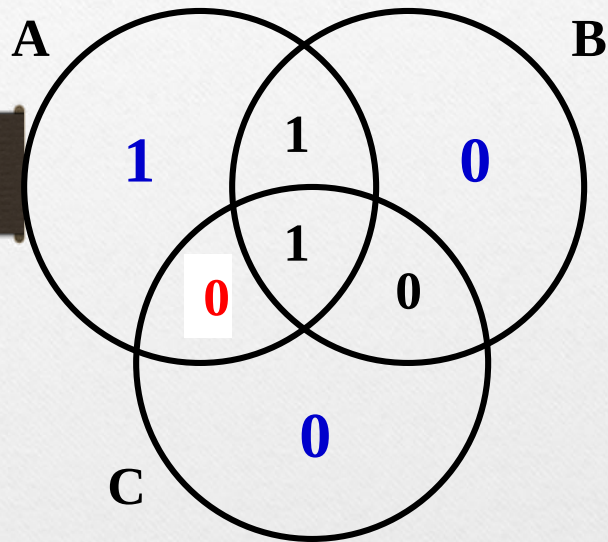
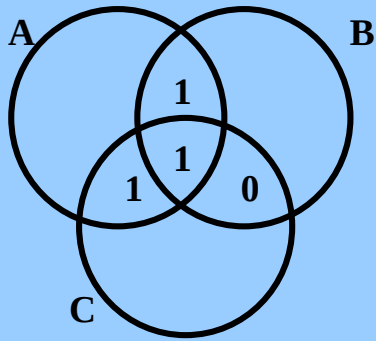


Fill the remaining compartments with parity bits.

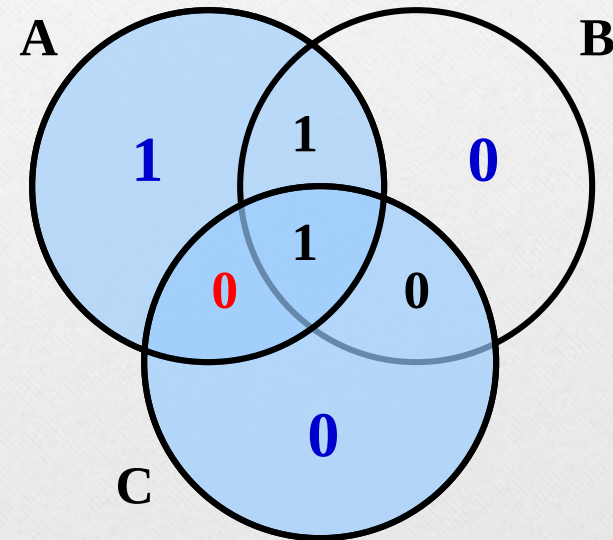
The total number of bits in a circle must be even.

For example: The data bits in A = $1+1+1 = 3$. This is odd – therefore add an additional 1.

Hamming Code



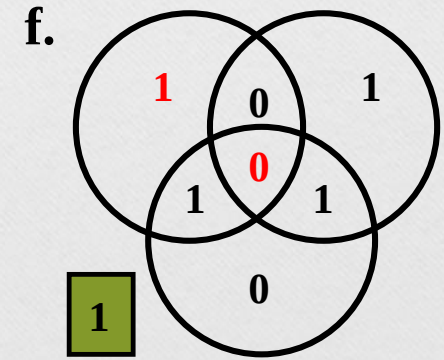
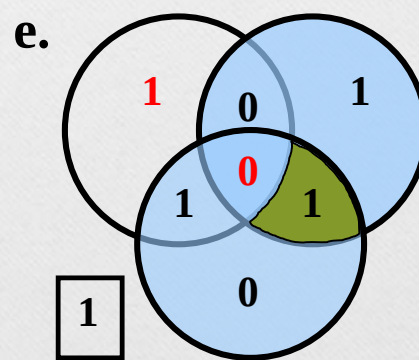
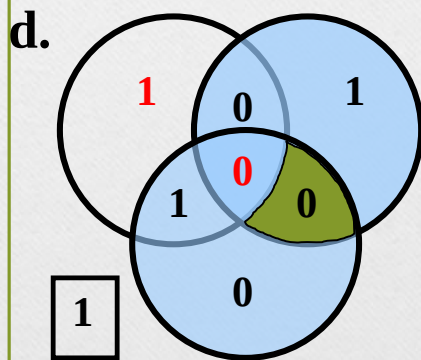
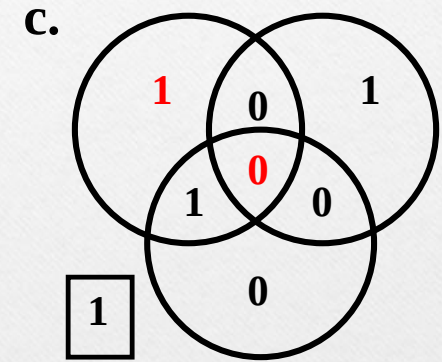
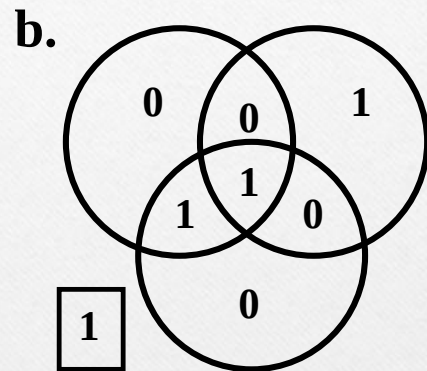
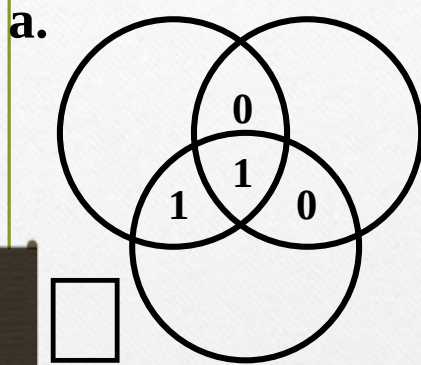
If a bit gets erroneously changed, the parity bits in that circle will no longer add up to 1.



Errors are found in A and C – and the shared bit in A and C is in error and can be fixed.

Double Error Detecting

- Previous example is Single-Error-Correcting code.
- Semiconductor memory is usually equipped with SEC-DED (Single-error-correcting, double-error-detecting code). SEC-DED requires an extra bit.



Single-Error –Correcting code

- 8 data bits
- $2^K - 1 \geq M + K$ is used to find the value of K (number of check bits).
- $2^K - 1 \geq 8 + K \Rightarrow K = 4$

Step1:

To find the number of check bits required for the given data bits.

Step2:

Store the data bits along with the check bits

Step3:

Retrieve the data and calculate the check bits again

Step4:

Generate the Syndrome word by comparing the two check bits

Step5:

- a. Syndrome contain all 0s no errors has been detected.
- b. If only one bit is set as 1 then an error has occurred in one of the check bits. No correction is required..
- c. If more than one bit is set to 1, then the numerical value of the syndrome indicate the position of the data bit in error. This data bit is inverted for correction.

Single Bit Errors in 8-bit words

- Data and check bits arranged into a 12-bit word.
- Bit positions numbered from 1 to 12.
- Bit positions representing position numbers that are powers of 2 are designated as check bits.
- Check bits calculated as follows:

$C1 = D1 \oplus D2 \oplus D4 \oplus D5 \oplus D7$
$C2 = D1 \oplus D3 \oplus D4 \oplus D6 \oplus D7$
$C4 = D2 \oplus D3 \oplus D4 \oplus D8$
$C8 = D5 \oplus D6 \oplus D7 \oplus D8$

- Data and check bits arranged into a 12 bit word:

8 data bits	4 check bits
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Calculating check bits

Bit Position	Binary	Type		
1	0001	C1		$\oplus$ All D's with a 1 in bit 1
2	0010	C2		$\oplus$ All D's with a 1 in bit 2
3	0011		D1	
4	0100	C4		$\oplus$ All D's with a 1 in bit 3
5	0101		D2	
6	0110		D3	
7	0111		D4	
8	1000	C8		$\oplus$ All D's with a 1 in bit 4
9	1001		D5	
10	1010		D6	
11	1011		D7	
12	1100		D8	

$$C1 = D1 \oplus D2 \oplus D4 \oplus D5 \oplus D7$$

Each check bit works on every data bit who shares the same bit position

Example

- Input word: 00111001
Data bit D1 in rightmost position

- Calculate check bits:

$$C1 = 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 = 1$$

$$C2 = 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 = 1$$

$$C4 = 0 \oplus 0 \oplus 1 \oplus 0 = 1$$

$$C8 = 1 \oplus 1 \oplus 0 \oplus 0 = 0$$

Stored word = 001101001111

- If data bit 3 sustains an error (001101101111)

$$C1 = 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 = 1$$

$$C2 = 1 \oplus 1 \oplus 1 \oplus 1 \oplus 0 = 0$$

$$C4 = 0 \oplus 1 \oplus 1 \oplus 0 = 0$$

$$C8 = 1 \oplus 1 \oplus 0 \oplus 0 = 0$$

- Calculate syndrome word:

0110 = bit position 6.

- D3 resides in bit position 6.

	C	C	C	C
	8	4	2	1
	0	1	1	1
$\oplus$	0	0	0	1
	0	1	1	0

3. Develop an SEC code for a 16-bit data word. Generate the code for the data word 0101000000111001. Show that the code will correctly identify an error in data bit 7. (23 points)

Solution:

Step 1 (2.5 points): According to the inequality $2^k - 1 \geq M + K$, where $M=16$, 5 check bits are needed for an SEC code for 16-bit data words.

Step 2 (5 points): The layout of data bits and check bits:

Bit Position	Position Number	Check Bits	Data Bits
21	10101		D16
20	10100		D15
19	10011		D14
18	10010		D13
17	10001		D12
16	10000	C16	

15	01111		D11
14	01110		D10
13	01101		D9
12	01100		D8
11	01011		D7
10	01010		D6
9	01001		D5
8	01000	C8	
7	00111		D4
6	00110		D3
5	00101		D2
4	00100	C4	
3	00011		D1
2	00010	C2	
1	00001	C1	

Step 3 (8 points): The check bits are calculated as in the class examples, for example,

$$C1 = D1 \oplus D2 \oplus D4 \oplus D5 \oplus D7 \oplus D9 \oplus D11 \oplus D12 \oplus D14 \oplus D16.$$

$$C16 = D12 \oplus D13 \oplus D14 \oplus D15 \oplus D16$$

For the word 0101000000111001, the check bits are
 $C_{16} = 0$; $C_8 = 0$; $C_4 = 0$; $C_2 = 0$; $C_1 = 1$.

Thus, the code word is (2.5 points):

01010000001101000101

If an error occurs in data bit 7:

$C_{16} = 0$; $C_8 = 1$; $C_4 = 0$; $C_2 = 1$; $C_1 = 0$.

Step 4 (5 points): Comparing the two sets of check bits forms the syndrome word:

	C16	C8	C4	C2	C1
	0	0	0	0	1
$\oplus$	0	1	0	1	0
	0	1	0	1	1

The result indicates an error identified in bit position 11, which is data bit 7

2. How many check bits are needed if the Hamming correcting code is used to detect single bit errors in a 512-bit data word? (7 points)

Solution:

Need K check bits such that: $512 + K \leq 2^K - 1$.

The minimum value of K , which satisfies this condition, is 10.

Problems

- Suppose an 8-bit data word stored in memory is 11000010. Using the Hamming algorithm, determine what check bits would be stored in memory with the data word. Show how you got your answer.
- How many check bits are needed if the hamming error correction code is used to detect single bit errors in 1024 bit data word
- Generate the code for the data word 0101000000111001. show that the code will correctly identify an error in data bit 5.

References

- William Stallings “Computer Organization and architecture” Prentice Hall, 7th edition, 2006
- Internet: PPT slides, Author: Jane and Harold Huang